Example 9.6.3

A bakery produces 6 kinds of pastry, including eclairs. Assume a supple of 20 each.

1. How many selections of 20 pastries? $N(T) = \binom{20+6-1}{20} = \binom{20}{20} = \frac{25!}{20!5!} = 53,130$
2. How many selections of 20 pastries including at least 3 eclairs? $\binom{17+6-1}{17} = \binom{22}{17} = \frac{22!}{17!5!} = 26,334$
3. How many selections of 20 pastries contain at most 2 eclairs? $N(E \leq 2) = N(T) - N(E \geq 3) = 53,130 - 26,334 = 26,796$

Example 9.6.9

```
for k = 1 to n
    for j = k to n
        for i = j to n
```

Number of iterations of inner loops is the same as the number of integers triples (i, j, k) where $1 \leq k \leq j \leq i \leq n$. Triples can be represented as a string of *n - 1* vertical bars and three crosses indicated which integers from *1* to *n* are included.
$\binom{3+n-1}{3} = \binom{n+2}{3} = \frac{n(n+1)(n+3)}{6}$

Example 9.6.12

$y_1 + y_2 + y_3 + y_4 = 30$. How many selections? $\binom{30+4-1}{30}$

Example 9.6.13

$y_1 + y_2 + y_3 + y_4 = 30, y_1 \geq 2, y_2 \geq 2, y_3 \geq 2, y_4 \geq 2.$ $\binom{22+4-1}{22}$

Example 9.6.18

A large pile of coins consist of pennies, nickels, dimes, and quarters.

- How many different collections of 30 coins can be selected? $N(T) = \binom{30+4-1}{30)} = \binom{33}{30} =$ 5,456.
- If the pile contains only 15 quarters, but 30 of each other kind of coin, how many collections of 30 can be chosen? $N(Q \geq 16) =$

  Let T be the set of selections of 30 coins, $Q \leq 15$ the set without most 15 quarters, and $Q \geq 6$, The set without at least 16 quarters. Then:

  $T = Q \leq 15 \cup Q \geq 16$
  $Q \leq 15 \cap Q \geq 16 = \emptyset$

$N(T) = N(Q \leq 15) + N(Q \geq 16) - \cancel{N}(Q \leq 15 \cap Q \geq 16)$
$N(Q \geq 16) = \binom{14+4-1}{14} = \binom{17}{14} = 680.$
$N(Q \leq 15) = 5,456 - 680 = 4,776$

- 20 Dimes, 30 of each of the rest. How many 30? T = $D \leq 20 \cup D \geq 21$
  Thus: $N(T) = N(D \leq 20) + N(D \geq 21)$
  $N(D \geq 21) = \binom{9+4-1}{9} = \binom{12}{9} = 220$
  $N(Dleq20) = N(T) - N(D \geq 21) = 5,456 - 220 = 5,236$

- 15 dimes, 15 quarters, 30 of each of the rest. How many 30? $N(Q \leq 15 \wedge D \leq 20) = N(T) - N(Q \geq 16 \cup D \geq 21)$
  $= 5,456 - 900 = 4,556$

  $N(Q \geq 16 \wedge D \geq 21) = N(Q \geq 16) + N(D \geq 21) - N(Q \geq 16 \cup D \geq 21)$
  $= 680 + 220 - 0 = 900$

9.1 Graph Theory

9.2 Graph Theory I